

Solutions by:
Mikhail Gaertan

Calculus for Biology and Medicine, 3rd Edition, Claudia Neuhauser

10.3 #1, 7, 16, 29, 30, 31, 33, 38, 39, 44, 48

1) $f(x, y) = x^2y + xy^2 \rightarrow f(x, y) = \cos^2(x^2 - 2y)$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2y + xy^2) = 2xy + y^2$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(\cos^2(x^2 - 2y)) = 2\cos(x^2 - 2y)\sin(x^2 - 2y)(2x)$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2y + xy^2) = x^2 + 2xy$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(\cos^2(x^2 - 2y)) = 2\cos(x^2 - 2y)\sin(x^2 - 2y)(-2)$$

16) $f(x, y) = \log_5(3xy)$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(\log_5(3xy)) = \frac{1}{\log 5} \cdot \frac{1}{3xy} \cdot 3y = \frac{1}{x \log 5}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(\log_5(3xy)) = \frac{1}{\log 5} \cdot \frac{1}{3xy} \cdot 3x = \frac{1}{y \log 5}$$

29) $P_e(a, T) = \frac{aNT}{1 + aT_hN}$

(a) $\frac{\partial P_e}{\partial a} = \frac{(1 + aT_hN)NT - aNT(T_hN)}{(1 + aT_hN)^2} = \frac{NT}{(1 + aT_hN)^2} > 0 \Rightarrow$

An increase in a will cause an increase in the prey eaten per predator.

(b) $\frac{\partial P_e}{\partial T} = \frac{aN}{1 + aT_hN} > 0 \Rightarrow$

An increase in T will cause an increase in the prey eaten per predator.

30) $\frac{1}{H} \frac{dH}{dt} = r \left(1 - \frac{H}{K} \right) - aP$ $\frac{1}{H} \frac{dH}{dt}$ = per capita growth rate of the prey

(a) $\frac{\partial}{\partial H} \left(\frac{1}{H} \frac{dH}{dt} \right) = \frac{\partial}{\partial H} \left(r \left(1 - \frac{H}{K} \right) - aP \right) = -\frac{r}{K} < 0 \Rightarrow$

An increase in prey density will cause a decrease in the per capita growth rate of the prey.

(b) $\frac{\partial}{\partial P} \left(\frac{1}{H} \frac{dH}{dt} \right) = \frac{\partial}{\partial P} \left(r \left(1 - \frac{H}{K} \right) - aP \right) = -a < 0 \Rightarrow$

An increase in predator density will cause a decrease in the per capita growth rate of the prey.

$$31) f(x, y, z) = x^2 z + y z^2 - xy$$

$$\frac{\partial f}{\partial x} = 2xz - y$$

$$\frac{\partial f}{\partial y} = z^2 - x$$

$$\frac{\partial f}{\partial z} = x^2 + 2yz$$

$$33) f(x, y, z) = x^3 y^2 z + \frac{x}{yz}$$

$$\frac{\partial f}{\partial x} = 3x^2 y^2 z + \frac{1}{yz}$$

$$\frac{\partial f}{\partial y} = 2x^3 y z - \frac{x}{y^2 z}$$

$$\frac{\partial f}{\partial z} = x^3 y^2 - \frac{x}{y z^2}$$

$$38) f(x, y, z) = y \tan(x^2 + z)$$

$$\frac{\partial f}{\partial x} = 2xy \sec^2(x^2 + z)$$

$$\frac{\partial f}{\partial y} = \tan(x^2 + z)$$

$$\frac{\partial f}{\partial z} = y \sec^2(x^2 + z)$$

$$39) f(x, y) = x^2 y + xy^2$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} (x^2 y + xy^2) \right)$$

$$= \frac{\partial}{\partial x} (2xy + y^2)$$

$$= 2y$$

$$44) g(s, t) = \ln(s^2 + 3st)$$

$$\frac{\partial^2 g}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial g}{\partial t} \right) = \frac{\partial}{\partial t} \left(\frac{\partial}{\partial t} (\ln(s^2 + 3st)) \right)$$

$$= \frac{\partial}{\partial t} \left(\frac{3s}{s^2 + 3st} \right)$$

$$= \frac{\partial}{\partial t} [3s(s^2 + 3st)^{-1}]$$

$$= -3s(3s)(s^2 + 3st)^{-2}$$

$$= \frac{-9s^2}{(s^2 + 3st)^2}$$

$$48) f(x, y) = \sin(3xy)$$

$$\frac{\partial^3 f}{\partial y^2 \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \right)$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (\sin(3xy))$$

$$= 3y \cos(3xy)$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (3y \cos(3xy))$$

$$= 3y(-\sin(3xy))(3x) + 3\cos(3xy) = 3\cos(3xy) - 9xy \sin(3xy)$$

$$\frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \right) = \frac{\partial}{\partial y} (3\cos(3xy) - 9xy \sin(3xy))$$

$$= 3(-\sin(3xy))3x - [9xy(\cos(3xy))(3x) + 9x \sin(3xy)]$$

$$= -27x^2 y \cos(3xy) - 18x \sin(3xy)$$